

Robotics Setter Technician Program

Robotic welding, from learning to improvement

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Engineering Department | August 1, 2026 to January 1, 2027

TENNECO

THE TENNECO VALUE I IDENTIFY WITH

Win

We must earn the trust of our employees and customers.

For me, winning meant taking the longest, hardest weld off a team member's hands.

Program scope and objective

Robotics Setter Technician | August 1, 2026 to January 1, 2027

OBJECTIVE

Become a specialist in robotic welding and advanced technical support.

WHY IT MATTERS

When a welding robot stops, the line stops with it. A setter who can find the cause and adjust the cell on the spot keeps parts moving and quality under control.

FINAL PROJECT

An implemented improvement

In quality, availability or productivity.

Mine: a manual weld moved to the 4M robotic cell.

An updated training process

How the next setters learn.

Mine: study material reviewed, schedule rebuilt.



Four levels, from understanding to improving

The skill path of the program

I

Understands

Safety, the welding process, the procedure (WPS), defects and settings.

L

Does it with support

Basic adjustments, checking settings and changing consumables.

U

Does it alone

Programming the robot (SKS), creating programs and setting them up.

O

Improves and teaches

Troubleshooting, training others, Lean and continuous improvement.

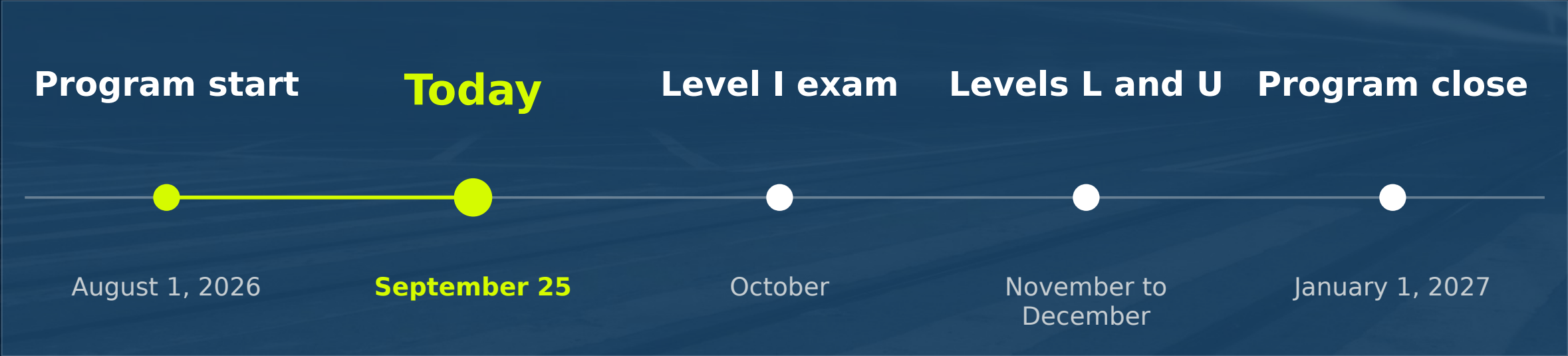
HOW IT WORKS

Each level builds on the one before: first understand, then do, then own it, then improve it.



Eight weeks in, fourteen to go

Program timeline



Level I: understand it before you touch the robot

What I learned

FIVE TOPICS

Safety	Working at the cell without putting anyone at risk.
GMAW welding	How wire, gas and an electric arc join the steel.
The WPS	The approved recipe that every weld has to follow.
Defects	Recognizing a bad weld and understanding its cause.
Parameters	The settings that shape the weld, and their limits.

HOW I LEARNED IT

616

pages of training manual and customer standard, read page by page.

Daily

at the cells, attending robot stops with the technicians.

25

questions of a practice exam from previous years, solved.

Four skills on my own, thirteen in progress

Progress on the 23 skills of the program

I Understands 9 skills

L With support 5 skills

U Alone 3 skills

O Improves 6 skills

On my own

In progress

Not started

ON MY OWN TODAY

- > Changing consumables
- > Adjusting the key welding settings
- > Reading weld symbols on drawings
- > Closing ANDON calls

Level I: exam still pending.



Most of my time goes to industrialization

How my time is split

60%

Industrialization project, with Engineering



40%

Support at the robotic cells: stops, adjustments and ANDON calls.

WHAT INDUSTRIALIZATION MEANS

Getting a new process ready to run on the floor: equipment, programs, trials and the release to production.

MY PART

[Project in one line: the product or line being industrialized.]

[What I do in it, in one or two short phrases.]

A long, uncomfortable weld, done by hand

Improvement | The starting point

THE STARTING POINT

One process ran in a manual welding booth.

➤ Time

Long welds made every part slow to finish.

➤ The team member

Uncomfortable positions, held for a long time, part after part.

The idea: let a robot do it.

PHOTO

Manual welding booth



From the manual booth to the 4M robotic cell

Improvement | What we did

01

Study the part

Which welds, how long they take, and what made them hard.



02

Move it to 4M

Bring the part into the automatic cell.



03

Program and adjust

Robot path and welding settings, until the weld is right.



04

Validate and release

Check weld quality, then hand it over to production.

TEAMWORK

Designed and implemented together with my supervisor, from the idea to the release.



Same part, less time, less strain

Improvement | Results

BEFORE | MANUAL BOOTH

XX min

per part, welded by hand

AFTER | 4M ROBOTIC CELL

XX min

per part, welded by the robot

XX%

less process time per part

XX

welds moved from the booth to the robot

XX min

of uncomfortable welding removed per shift



What I would change in how setters learn

Final project | Training process update, a proposal

01

Reviewed

616 pages of training manual and customer standard, page by page, against what Level I asks.

02

Found

Gaps: topics the exam asks about that the material barely covers, and an error in one key table.

03

Rebuilt

The schedule, in the plant's own format, so plan and reality can be compared week by week.

PROPOSAL

Use the corrected material and the new schedule with the next group of setters.



Next: from knowing it to doing it alone

What comes next, until January

NEXT LEVELS

L Validate settings and find defects at the cell, with support.

November

U Create and set up robot programs on my own (SKS).

December

O Solve problems, share what I learn, keep improving.

Already started with 4M

WHAT I NEED

- A date for my Level I exam
- Supervised time programming robots in SKS
- Your feedback on the training proposal



Conclusions and next steps

What is in place, and what comes next

IN PLACE

Level I studied in full, and four skills done on my own.

A manual weld moved to the 4M robotic cell: less time, less strain.

A reviewed study path, and a schedule that shows plan versus reality.

NEXT

Pass the Level I exam in October.

Reach Levels L and U before January.

Keep measuring 4M, and find the next weld to automate.



I came to learn robotic welding, and I am already improving it.

The background of the slide is a dark blue image of a tunnel. Multiple bright, white light trails streak across the frame from the top left towards the bottom right, creating a sense of motion and depth. The perspective of the tunnel walls and floor lines converges towards the right side of the image.

TENNECO

Thank you

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